

OpenSSL:

Encrypt and decrypt a message using OpenSSL.

♦ Main Difference – When to Use What?

Feature	AES	RSA
Type	Symmetric	Asymmetric
Keys	One key (same for encryption & decryption)	Two keys (Public & Private)
Speed	Fast (good for large data)	Slow (good for small data like keys)
Use Case	Encrypting files, disks, databases	Encrypting small data, key exchange

```
Command Prompt
Volume in drive C has no label.
Volume Serial Number is 4039-7C4F

Directory of C:\Users\HP

File Not Found

C:\Users\HP>dir message.txt
Volume in drive C has no label.
Volume Serial Number is 4039-7C4F

Directory of C:\Users\HP

01/24/2025  12:09 AM                22 message.txt
             1 File(s)                22 bytes
             0 Dir(s)  31,321,313,280 bytes free

C:\Users\HP>rename message.txt message.txt

C:\Users\HP>openssl enc -aes-256-cbc -salt -in message.txt -out encrypted.txt
enter AES-256-CBC encryption password:

Verifying - enter AES-256-CBC encryption password:

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

C:\Users\HP>openssl enc -d -aes-256-cbc -in encrypted.txt -out decrypted.txt
enter AES-256-CBC decryption password:

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

C:\Users\HP>
```

Activate Windows
Go to Settings to activate Windows.

80°F Mostly cloudy 7:27 PM 2/10/2025